A PROJECT Title

ON

"Real-Time Hospital Bed Management Dashboard"

Insem-End project submitted in partial fulfilment of the requirements for the Course

Front End Development Frameworks ***for the degree of***

BACHELOR OF TECHNOLOGY

IN

Computer Science and Engineering

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BY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Name of Title: Real-Time Hospital Bed Management Dashboard

1. Abstract:

The Hospital Bed Management Dashboard is a comprehensive web-based solution designed to improve the efficiency of hospital resource utilization by providing real-time visibility into bed availability across multiple wards. In many healthcare institutions, managing bed allocation manually can lead to delays, overcrowding, and inefficient patient handling. This project addresses these challenges by offering an interactive and centralized dashboard that tracks admissions, discharges, transfers, and maintenance status of beds.

The system enables hospital staff to monitor bed occupancy dynamically, ensuring that patients are assigned beds quickly and appropriately. It includes features such as ward-wise filtering, a visual bed map for easy identification of available and occupied beds, and forms for patient admission and discharge processes. Additionally, the transfer module facilitates seamless movement of patients between wards while maintaining accurate records. Beds marked under maintenance are automatically excluded from allocation, preventing operational conflicts.

Another important feature is the alert system, which notifies staff when a ward is nearing full capacity. This helps hospitals respond faster and make better decisions during busy periods. The system also keeps a record of all activities, such as admissions and transfers, making it easier to review past data when needed. Charts and visual summaries provide useful insights into how beds are being used over time.

Developed using modern web technologies including React.js for the frontend and Node.js with Express for the backend, the application ensures scalability and responsiveness. Optional integration with a MySQL database enables persistent data storage, while REST APIs facilitate smooth communication between components.

Overall, the Hospital Bed Management Dashboard enhances operational efficiency, reduces manual errors, and supports better patient care by streamlining hospital bed allocation and monitoring processes in a user-friendly and reliable manner.

2. INTRODUCTION:

Managing hospital beds efficiently is important for providing timely and effective patient care. In many hospitals, bed allocation is still handled manually, which can lead to delays, confusion, and inefficient use of resources. The Hospital Bed Management Dashboard provides a real-time view of bed availability across different wards, helping staff easily track which beds are available, occupied, or under maintenance. It also supports essential operations such as patient admission, discharge, and transfers between wards. By using a simple and centralized digital interface, the system reduces manual effort and minimizes errors. Overall, it improves efficiency and helps hospitals manage their resources in a more organized and effective way.

3. SOFTWARE REQUIREMENTS:

Frontend: HTML5, CSS3, JavaScript (ES6+), React.js, Bootstrap
Backend: Node.js, Express.js
Database: MySQL
Tools: npm, Postman (API testing), Visual Studio Code, Web Browser (Chrome/Edge)
Libraries: Axios, React Router, Context API / Redux
Storage: LocalStorage (for temporary data handling)

Course Coordinator **Course Instructor** **FED , Coordinator**